

Lecture Notes VI

Course: Functional Analysis

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Topic: Product Spaces

Proof 1.45 (Cross sections preserve set measurability). Suppose $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ are measurable spaces, then define $\mathcal{E}$ as the collection

$$\mathcal{E} = \{E \subseteq X \times Y : [E]_a \in \mathcal{N}, [E]^b \in \mathcal{M} \text{ for all } a \in X, b \in Y\}.$$

First, note that for any $A \subseteq X, B \subseteq Y$ we have

$$[A \times B]_a = \begin{cases} B & a \in A \\ \emptyset & a \notin A \end{cases} \quad \text{and} \quad [A \times B]^b = \begin{cases} A & b \in B \\ \emptyset & b \notin B \end{cases},$$

so that if A and B are both measurable, then so are the cross-sections of their Cartesian products, i.e., both $[A \times B]_a$ and $[A \times B]^b$ are in $\mathcal{E}$ for all $a \in X, b \in Y$ provided that A, B are both measurable. Furthermore, we have

$$[X \times Y \setminus E]_a = \{y \in Y : (a, y) \notin E\} = \{y \in Y : y \notin [E]_a\} = Y \setminus [E]_a,$$

and similarly, $[X \times Y \setminus E]^b = X \setminus [E]^b$. For unions we see that

$$\left[\bigcup_{n \in \mathbb{N}} E_n \right]_a = \{y \in Y : (a, y) \in [E_n]_a \text{ for some } n \in \mathbb{N}\} = \bigcup_{n \in \mathbb{N}} \{y \in Y : (a, y) \in [E_n]_a\} = \bigcup_{n \in \mathbb{N}} [E_n]_a.$$

It is now readily seen that $\mathcal{E}$ is a σ -algebra which contains any set of the form $A \times B$ with A, B measurable, which implies that $\mathcal{M} \otimes \mathcal{N} \subset \mathcal{E}$ (because $\mathcal{M} \otimes \mathcal{N}$ is the smallest σ -algebra containing the measurable rectangles).

Proof 1.46 (Cross sections preserve function measurability). Let $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ be measurable spaces and let $f : X \times Y \rightarrow \mathbb{R}$ be a measurable function. Then

$$y \in [f]_a^{-1}(D) \iff [f]_a(y) \in D \iff f(a, y) \in D \iff (a, y) \in f^{-1}(D) \iff y \in [f^{-1}(D)]_a$$

meaning that $[f]_a^{-1}(D) = [f^{-1}(D)]_a$, and thus if D is a Borel set, then $f^{-1}(D)$ is $\mathcal{M} \otimes \mathcal{N}$ -measurable and thus $[f^{-1}(D)]_a = [f]_a^{-1}(D)$ is $\mathcal{N}$ -measurable, and the same goes for the cross sections on b with $b \in Y$.

Proof 1.47 (Algebras on σ -algebras). Part i) readily follows from the definition of σ -algebra, and also holds for intersections (see the remark following the definition of σ -algebra). To prove ii) note that we have the decomposition

$$(A \times B) \cup (C \times D) = (A \times B) \cup (C \times (D \setminus B)) \cup ((C \setminus A) \times (B \cap D))$$

which writes a union of two measurable rectangles as a disjoint union of three disjoint measurable rectangles.

Proof 1.48 (Monotone Class Theorem). It is clear that any σ -algebra is a monotone class, so $\mathcal{M}(\mathcal{A}) \subseteq \sigma(\mathcal{A})$. Conversely, we begin by assuming that $A \in \mathcal{A}$ and letting

$$\mathcal{L} = \{E \in \mathcal{M}(\mathcal{A}) : A \cup E \in \mathcal{M}(\mathcal{A})\}$$

we immediately see that $\mathcal{A} \subseteq \mathcal{L}$. Also, since $\mathcal{M}(\mathcal{A})$ is a monotone class, it is readily seen that $\mathcal{L}$ is also a monotone class and hence $\mathcal{M}(\mathcal{A}) \subseteq \mathcal{L}$. We proved that for $A \in \mathcal{A}$ we have $A \cup E \in \mathcal{M}(\mathcal{A})$ for any $E \in \mathcal{M}(\mathcal{A})$ so that if we define

$$\mathcal{D} = \{D \in \mathcal{M}(\mathcal{A}) : D \cup E \in \mathcal{M}(\mathcal{A}) \text{ for all } E \in \mathcal{M}(\mathcal{A})\}$$

the previous result can be restated as $\mathcal{A} \subseteq \mathcal{D}$, but again, notice that $\mathcal{D}$ is a monotone class and thus $\mathcal{M}(\mathcal{A}) \subseteq \mathcal{D}$ so $D \cup E \in \mathcal{M}(\mathcal{A})$ for all $D, E \in \mathcal{M}(\mathcal{A})$ (*). Now if (E_n) is a sequence in $\mathcal{M}(\mathcal{A})$, then we can construct the (increasing) sequence (I_n) as

$$I_n = \bigcup_{k=1}^n E_k \implies \bigcup_{n \in \mathbb{N}} I_n = \bigcup_{n \in \mathbb{N}} E_n$$

and by (*) we have that $I_n \in \mathcal{M}(\mathcal{A})$ for any $n \in \mathbb{N}$ so $\mathcal{M}(\mathcal{A})$ is closed under countable unions. Finally, we define

$$\mathcal{M}(\mathcal{A})^\sim = \{E \in \mathcal{M}(\mathcal{A}) : W \setminus E \in \mathcal{M}(\mathcal{A})\}$$

we immediately get $\mathcal{A} \subseteq \mathcal{M}(\mathcal{A})^\sim$ because $\mathcal{A}$ is closed under complementation and since $\mathcal{M}(\mathcal{A})^\sim$ is also a monotone class, this means that $\mathcal{M}(\mathcal{A}) \subseteq \mathcal{M}(\mathcal{A})^\sim$ so $\mathcal{M}(\mathcal{A})$ is closed under complementation. We therefore conclude that $\mathcal{M}(\mathcal{A})$ is a σ -algebra containing $\mathcal{A}$ so $\sigma(\mathcal{A}) \subseteq \mathcal{M}(\mathcal{A})$ and the proof is done.

Proof 1.49 (Measure of cross sections are measurable functions). ii) We know by Theorem 1.45 that if $E \in \mathcal{M} \otimes \mathcal{N}$ then $[E]_y \in \mathcal{M}$ for any $y \in Y$ so that $\mu([E]^y)$ is well-defined. Let $\mathcal{R}$ be the collection of all measurable rectangles, i.e., elements of the form $A \times B$ with $A \in \mathcal{M}$ and $B \in \mathcal{N}$ and define

$$\mathcal{V} = \{E \in \mathcal{M} \otimes \mathcal{N} : g(y) = \mu([E]^y) \text{ is a measurable function on } Y\}$$

then clearly $\mathcal{V} \subseteq \mathcal{M} \otimes \mathcal{N}$. Moreover, given any measurable rectangle $A \times B$ in $\mathcal{R}$ we have that $g(y) = \mu([A \times B]^y) = \chi_B(y)\mu(A)$ for any $y \in Y$ and hence $g(y)|_{\mathcal{M} \otimes \mathcal{N}}$ is measurable on Y , i.e., $\mathcal{V}$ contains all measurable rectangles in $\mathcal{M} \otimes \mathcal{N}$, $\mathcal{R} \subseteq \mathcal{V}$. Next, let

$$\mathcal{A} = \left\{ \bigcup \mathcal{R} : \mathcal{R} \subseteq \mathcal{R} \text{ and } \mathcal{R} \text{ is finite} \right\},$$

which is an algebra on $X \times Y$ by Theorem 1.47.

- $\mathcal{A} \subseteq \mathcal{V}$: Suppose $E \in \mathcal{A}$, then by Theorem 1.47 we have that E is a union of disjoint measurable rectangles $E_1, \dots, E_n$ and thus

$$\mu([E]^y) = \mu\left(\left[\bigcup_{k=1}^n E_k\right]^y\right) = \mu\left(\bigcup_{k=1}^n [E_k]^y\right) = \sum_{k=1}^n \mu([E_k]^y)$$

so that $\mu([E]^y)$ is a finite sum of measurable functions (each $\mu([E_k]^y)$ is measurable because the rectangles $E_1, \dots, E_n$ in the decomposition are measurable) and thus $E \in \mathcal{V}$

- $\mathcal{V}$ is a monotone class on $X \times Y$: Given any increasing sequence (E_n) in $\mathcal{V}$ we have

$$\mu\left(\left[\bigcup_{n \in \mathbb{N}} E_n\right]^y\right) = \mu\left(\bigcup_{n \in \mathbb{N}} [E_n]^y\right) = \lim \mu([E_n]^y),$$

the last equality following from continuity from below of a measure of Theorem 1.23. We got that the measure of the countable union is the (pointwise) limit of measurable functions (each function is measurable because (E_n) is a sequence in $\mathcal{V} \subseteq \mathcal{M} \otimes \mathcal{N}$) and thus it is measurable (pointwise limits preserve measurability) and hence the countable union of E_n 's is in $\mathcal{V}$. Further, given any decreasing sequence (E_n) in $\mathcal{V}$ we have

$$\mu\left(\left[\bigcap_{n \in \mathbb{N}} E_n\right]^y\right) = \mu\left(\bigcap_{n \in \mathbb{N}} [E_n]^y\right) = \lim \mu([E_n]^y)$$

only if $\mu(E_1) < \infty$, in order to ensure such condition, let us assume that μ is a finite measure[†]. Then by the same reasoning as before, we conclude that $\mathcal{V}$ is closed under countable intersections of decreasing sequences. We therefore conclude that $\mathcal{V}$ is a monotone class on $X \times Y$.

The previous results show that $\mathcal{V}$ is a monotone class on $X \times Y$ which contains the algebra $\mathcal{A}$ meaning that $\mathcal{M}(\mathcal{A}) \subseteq \mathcal{V}$ but by Theorem 1.48 (Monotone Class Theorem) we have that $\sigma(\mathcal{A}) = \mathcal{M}(\mathcal{A}) \subseteq \mathcal{V}$ so $\sigma(\mathcal{A}) \subseteq \mathcal{V}$ but σ -algebras are closed under finite unions so $\mathcal{M} \otimes \mathcal{N} = \sigma(\mathcal{R}) = \sigma(\mathcal{A})$ and hence $\mathcal{M} \otimes \mathcal{N} \subseteq \mathcal{V}$. We therefore conclude that $\mathcal{M} \otimes \mathcal{N} = \mathcal{V}$ but only for finite measures ($\dagger$) so to extend the argument let μ be a σ -finite measure, meaning that there exists a sequence (X_n) of measurable sets with finite measure such that their union is X , in particular, if we define the sequence (I_n) by $I_n = \bigcup_{k=1}^n X_k$, then clearly $\bigcup_{n \in \mathbb{N}} X_n = X$, and we still have $\mu(I_n) < \infty$ for each n , but now (I_n) is an increasing union meaning that for any $E \in \mathcal{M} \otimes \mathcal{N}$ we have

$$g(y) = \mu([E]^y) = \lim \mu([E \cap (X_n \times Y)]^y) = \lim \mu([E]^y \cap X_n)$$

now, for any given X_n define $\mathcal{M}_n = \mathcal{M} \cap \mathcal{P}(X_n)$ which is a σ -algebra on X_n (in this sense the measurable space is $(X_n, \mathcal{M}_n)$) so that $\mu|_{\mathcal{M}_n}$ is finite and thus measurable by the previous proof, so that $\mu([E]^y \cap X_n) = \mu|_{\mathcal{M}_n}([E]^y \cap X_n)$, and thus we have that $g(y) = \mu([E]^y)$ is the pointwise limit of a sequence of measurable functions, i.e., $g(y)$ is a measurable function on Y , completing the proof of ii). The proof of i) is done in a similar fashion.

Proof 1.51 (Characteristic function of a measurable rectangle is the characteristic function of a cross section). The lemma is an immediate consequence of:

$$(x, y) \in E \iff y \in [E]_x \text{ for any } x \in X, \quad \text{and} \quad (x, y) \in E \iff x \in [E]^y \text{ for any } y \in Y.$$

Proof 1.53 (Product measure is indeed a measure). The emptiness property is trivial so let (E_n) be a sequence of $\mathcal{M} \otimes \mathcal{N}$ -measurable sets, then

$$\begin{aligned} (\mu \otimes \nu) \left(\prod_{n \in \mathbb{N}} E_n \right) &= \int_X \int_Y d\nu(y) d\mu(x) \chi_{\prod_n E_n}(x, y) = \int_X \int_Y d\nu(y) d\mu(x) \chi_{[\prod_n E_n]_x}(y) \\ &= \int_X d\mu(x) \nu \left(\left[\prod_{n \in \mathbb{N}} E_n \right]_x \right) = \int_X d\mu(x) \nu \left(\prod_{n \in \mathbb{N}} [E_n]_x \right) = \int_X d\mu \sum_{n \in \mathbb{N}} \nu([E_n]_x) = \sum_{n \in \mathbb{N}} \int_X d\mu(x) \nu([E_n]_x) \\ &= \sum_{n \in \mathbb{N}} \int_X \int_Y d\nu(y) d\mu(x) \chi_{[E_n]_x}(y) = \sum_{n \in \mathbb{N}} \int_X \int_Y d\nu(y) d\mu(x) \chi_{E_n}(x, y) = \sum_{n \in \mathbb{N}} (\mu \otimes \nu)(E_n), \end{aligned}$$

The second equality is just Lemma 1.51 and the interchange of integral with the series is justified because $\nu([E_n]_x) \geq 0$ for any $n \in \mathbb{N}$ so that the series is increasing and thus Theorem 1.37 (Beppo Levi's MCT) applies. Thus $\mu \otimes \nu$ is a measure on $(X \times Y, \mathcal{M} \otimes \mathcal{N})$.

Proof 1.54 (Tonelli's Theorem). First, consider the case of a characteristic functions, i.e., let $f = \chi_E$ for some $E \in \mathcal{M} \otimes \mathcal{N}$. In this case

$$I_Y(x) = \int_Y d\nu \chi_E = \nu([E]_x) \quad \text{for every } x \in X$$

and

$$I_X(y) = \int_X d\mu \chi_E = \mu([E]^y) \quad \text{for every } y \in Y$$

so that I_Y, I_X are both measurable by Theorem 1.49. Next, for the commutativity of iteration, let us define

$$\mathcal{D} = \left\{ E \in \mathcal{M} \otimes \mathcal{N} : \int_X \int_Y d\nu d\mu \chi_E = \int_Y \int_X d\mu d\nu \chi_E \right\} \quad \text{and} \quad \mathcal{R} = \{A \times B : A \in \mathcal{M}, B \in \mathcal{N}\}.$$

Given any $A \times B \in \mathcal{R}$ we have that

$$\int_X \int_Y d\nu d\mu \chi_{A \times B}(x, y) = \int_X \int_Y d\nu d\mu \chi_{[A \times B]_x}(y) = \int_X d\mu \nu([A \times B]_x) = \int_X d\mu \nu(B) \chi_A(x) = \nu(B) \mu(A)$$

and

$$\int_Y \int_X d\mu d\nu \chi_{A \times B}(x, y) = \int_Y \int_X d\mu d\nu \chi_{[A \times B]^y}(x) = \int_Y d\nu \mu([A \times B]^y) = \int_Y d\nu \mu(A) \chi_B(y) = \mu(A) \nu(B)$$

so that $\mathcal{R} \subseteq \mathcal{D}$. Next, let

$$\mathcal{A} = \left\{ \bigcup \mathcal{F} : \mathcal{F} \subseteq \mathcal{R} \text{ and } \mathcal{F} \text{ is finite} \right\}$$

so now Theorem 1.47 implies that any element of $\mathcal{A}$ can be written as a disjoint union of elements in $\mathcal{R}$, and using $\chi_A = \chi_{\bigsqcup_{k=1}^n A_k} = \sum_{k=1}^n \chi_{A_k}$, the fact that the integral is linear and $\mathcal{R} \subseteq \mathcal{D}$ we have that $\mathcal{A} \subseteq \mathcal{D}$. Furthermore, if (E_n) is any increasing sequence in $\mathcal{D}$ we have that

$$\begin{aligned} \int_X \int_Y d\nu d\mu \chi_{\bigcup_n E_n} &= \int_X d\mu \nu \left(\bigcup_{n \in \mathbb{N}} [E_n]_x \right) = \int_X d\mu \lim \nu([E_n]_x) = \lim \int_X d\mu \nu([E_n]_x) = \lim \int_X \int_Y d\nu d\mu \chi_{E_n} \\ &= \lim \int_Y \int_X d\mu d\nu \chi_{E_n} = \int_Y d\nu \mu([E_n]^y) = \int_Y d\nu \lim \mu([E_n]^y) = \int_Y d\nu \mu \left(\bigcup_{n \in \mathbb{N}} [E_n]^y \right) = \int_Y \int_X d\mu d\nu \chi_{\bigcup_n E_n}, \end{aligned}$$

where the first appearance of the limit is just continuity from below of a measure and the interchange of limit and integral is justified because (E_n) is increasing, meaning that $E_n \subseteq E_m$ for $m \geq n$ so the sequences $(\nu([E_n]_x))$, $(\mu([E_n]^y))$ are both increasing by monotonicity of measures and nonnegative by definition of positive measure. Thus $\mathcal{D}$ is closed under countable increasing unions, and similarly for countable decreasing intersections, but in that case the interchange of limits cannot be argued as in the latter case because the inequality goes the other way around, but since the sequence is decreasing, we have that $\mu([E_n]^y) \leq \mu([E_1]^y)$ for all $n \in \mathbb{N}$ (and similarly for ν) so let us assume that μ, ν are finite measures so now Theorem 1.44 (Lebesgue's DCT) with the dominating function being a constant ($\mu([E_1]^y), \nu([E_1]_x)$, respectively) gives the justification for the interchange of the integral with the limit.

$\therefore \mathcal{D}$ is a monotone class.

We have shown that $\mathcal{D}$ is a monotone class that contains the algebra $\mathcal{A}$ so that $\mathcal{M}(\mathcal{A}) \subseteq \mathcal{D}$ and by the monotone class theorem along with the fact that $\sigma(\mathcal{A}) = \sigma(\mathcal{R})$ we get that $\mathcal{M} \otimes \mathcal{N} \subseteq \mathcal{D}$.

$$\therefore \int_X \int_Y d\nu d\mu \chi_E = \int_Y \int_X d\nu d\mu \chi_E \text{ for all } E \in \mathcal{M} \otimes \mathcal{N} \text{ and } \mu, \nu \text{ finite measures.}$$

Now relax the assumption that μ, ν are finite and let them be σ -finite so that (see Problem #.#) there exist increasing sequences $(X_n), (Y_n)$ in $\mathcal{M}, \mathcal{N}$, respectively, such that each set has finite measure and such that X, Y are the union of such sets, respectively. For any $E \in \mathcal{M} \otimes \mathcal{N}$ we must have that the assertion holds true —using the restriction of the measures just as in **Proof 1.49**— for $E \cap [X_j \times Y_k]$ with j, k any natural numbers so for any fixed k we can let j run and by Theorem 1.37 (Beppo Levi's MCT) we get that the assertion holds true for $E \cap [X \times Y_k]$, now let k run and appealing to the same reasoning we conclude that the assertion holds true for $E \cap [X \times Y] = E$.

$$\therefore \int_X \int_Y d\nu d\mu \chi_E = \int_Y \int_X d\nu d\mu \chi_E \text{ for all } E \in \mathcal{M} \otimes \mathcal{N} \text{ and } \mu, \nu \text{ } \sigma\text{-finite measures.}$$

Note that the measure of E wrt $\mu \otimes \nu$ is just the first integral in the conclusion so that last assertion of the theorem holds for characteristic functions of any measurable set $E \in \mathcal{M} \otimes \mathcal{N}$. The extension to general functions is easily done now using Lemma 1.33 (Approximation Lemma) —recalling that the functions under consideration are nonnegative— and Theorem 1.37 (Beppo Levi's MCT). The proof is done.